

STAR CLUSTERS

More than 2,000 open clusters have been cataloged in the Milky Way. About half contain fewer than 100 stars, but the largest have more than 1,000. Open clusters are asymmetrical and range in size from 5 to 75 light-years across. By contrast, globular clusters contain up to a million stars, spread symmetrically across several hundred light-years. Only about 150 globular clusters are known in the Milky Way and, unlike open clusters, which are found mainly in the galaxy's spiral arms, most are scattered around the periphery.

MASSIVE CLUSTER

Omega Centauri is a prime example of a globular star cluster. It contains more than 10 million old stars and has a mass of 5 million solar masses.

OPEN CLUSTER

Hyades



TAURUS

CATALOG NUMBER
MEL 25

DISTANCE FROM SUN
150 light-years

MAGNITUDE 0.5

The Hyades is one of the closest open clusters to Earth and has been recognized since ancient times. The brightest of the cluster's 200 stars form a V-shape in the sky, clearly visible to the naked eye. The cluster's central group is about 10 light-years

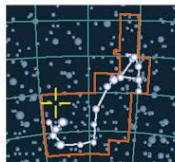
in diameter, and its outlying members span up to 80 light-years. Most of the stars in this cluster are of spectral classes G and K (see pp.232–233) and are average in size, with temperatures comparable to that of the Sun. The brightest star in the field of the Hyades, the red giant Aldebaran (see p.256), is not a member of the cluster and is much closer to Earth. The cluster's stars all move in a common direction, toward a point east of Betelgeuse in Orion (see p.256). Studies of the movement of the stars of the Hyades show that they have a common origin with the Beehive Cluster (see below). The Hyades is thought to be about 790 million years old, and this age matches that of the Beehive Cluster. The parallel movement of stars in the Hyades has allowed their distance to be measured using the moving cluster method for stellar distances (see pp.232–233).

PROMINENT CLUSTER

First recorded by Homer around 750 BCE, the Hyades is one of the few star clusters visible to the naked eye. Aldebaran, the bright red giant in this image, is not part of the cluster but is 90 light-years closer to Earth.

OPEN CLUSTER

Butterfly Cluster



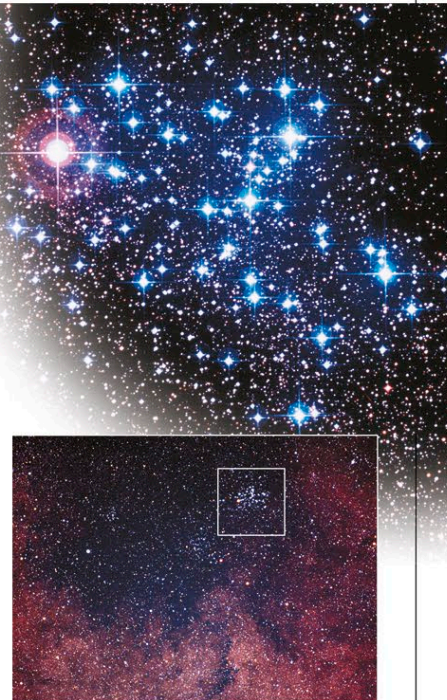
SCORPIUS

CATALOG NUMBERS
M6, NGC 6405

DISTANCE FROM SUN
2,000 light-years

MAGNITUDE 5.3

The Butterfly Cluster, located toward the center of the Milky Way, is about 12 light-years across and has an estimated age of 100 million years. In the night sky, the cluster occupies an area the size of a full moon, and, to some, it resembles the shape of a butterfly. The cluster is made up of about 80 stars, most of them very hot, blue main-sequence stars with spectral types B4 and B5 (see pp.232–233). The brightest star in the cluster, BM Scorpii, is an orange supergiant star that is also a semi-regular variable (see pp.282–283). At its brightest, this star is visible to the naked eye; at its faintest, binoculars are needed. The Butterfly Cluster displays a striking contrast between the blue main-sequence stars and the orange supergiant.



SKY SPECTACLE

The Butterfly Cluster is one of the largest and brightest open star clusters in the Milky Way. It can best be seen with binoculars in a dark sky and can be located within the constellation Scorpius.

OPEN CLUSTER

Beehive Cluster



CANCER

CATALOG NUMBER
M44

DISTANCE FROM SUN
577 light-years

MAGNITUDE 3.7

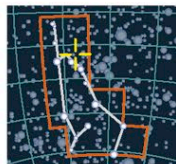
The Beehive Cluster, also known as Praesepe, is easily visible to the naked eye. The cluster contains over 350 stars spread across 10 light-years, but most of them can be seen only with a large telescope. It is thought to be about 730 million years old. Age, distance, and motion measurements

suggest that the Beehive Cluster most likely originated in the same star-forming nebula as the Hyades (above).

CELESTIAL BEEHIVE

OPEN CLUSTER

M93



PUPPIS

CATALOG NUMBERS
M93, NGC 2447

DISTANCE FROM SUN
3,600 light-years

MAGNITUDE 6

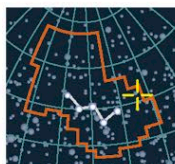
M93 is a bright open cluster and, at about 25 light-years across, is relatively small. It lies in the southern sky, close to the galactic equator. The cluster consists of about 80 stars, but only a few of the stars—blue giants of spectral type B9 (see pp.232–233)—provide most of the cluster's light. At about 100 million years old, M93 is young in astronomical terms.

SOUTHERN CLUSTER



OPEN CLUSTER

M52



CASSIOPEIA

CATALOG NUMBERS
M52, NGC 7654

DISTANCE FROM SUN
3,000–7,000 light-years

MAGNITUDE 7.5

An open cluster of about 200 stars, M52 lies against a rich Milky Way background. It was first cataloged in 1774 by Charles Messier (see p.73). The distance to the cluster is uncertain, with estimates ranging from 3,000 to 7,000 light-years. The uncertainty is due to high interstellar absorption that affects the light from the cluster during its journey to Earth. The uncertain distance also means that the cluster's size is unknown, but midrange estimates give a size of about 20 light-years across. The age of the cluster is calculated to be about 35 million years. The brightest stars in M52 have magnitudes of only 7.7 and



CLUSTER AND NEBULA

This image, stretching more than twice the diameter of a full moon, captures the open cluster M52 (top left) and the glowing Bubble Nebula (bottom right).

8.2, and with an overall magnitude of 7.5, the cluster is too faint to be seen with the naked eye. However, through binoculars, the cluster can be viewed as a faint nebulous patch, while a small telescope reveals a rich, compressed cluster of stars.